

RUSHIL VENKATESWAR

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EDUCATION

Indian Institute of Technology, Kharagpur

Dec 2020 – May 2025

Integrated Bachelors and Masters of Technology in Computer Science

8.76 / 10

Little Flower School

Apr 2018 – Apr 2020

Indian School Certificate Examination (ISC)

96.25%

AWARDS AND ACHIEVEMENTS

- Achieved a peak rating of **1576 (Specialist)** on **Codeforces** and **1868 (Knight)** on **LeetCode** under the handle **rv4102**
- Secured an **AIR 473** in Joint Entrance Exam Advanced 2020 and an **AIR 1176** in Joint Entrance Exam Mains 2020
- Nominated for **Best Bachelor Thesis Award** by Department of Computer Science and Engineering, IIT Kharagpur
- Selected to attend **Optiver Winter School**, a workshop on Quantitative Trading, conducted by IIT Delhi

EXPERIENCE

Sprinklr, India | *Product Engineering Intern* | *Workflow Automation*

May 2024 – July 2024

Objective: Automate after-call workflow graph creation using a chatbot interface and guided-path graph summarization using LLMs

- Ensured **structured output generation** from LLMs using **Pydantic** class definitions, allowing **data validation & serialization**
- Utilized **prompt engineering, function calling** & query re-asking through Instructor package to generate workflows from queries
- Modified open-source packages for compatibility, **dockerizing** and deploying the tools to reduce workflow creation time by over **60%**

Stanford University | *Research Intern* | *Maternal and Child Health Monitoring in LMICs*

May 2023 – Aug 2023

Objective: Estimate 6 key indicators of health status in low income countries using numeric features derived from satellite imagery

- Trained a **Boosting on Error** ensemble to predict 4 level residuals using **Random Forest & XGBoost**, with MCRMSE of 10.759
- Utilized **Dask** package on an **8GB** dataset for **EDA**, finally using `RF.feature_importances_` parameter to perform feature selection
- The internship program was conducted as a Kaggle Competition, and our team secured **1st position** out of 30 participating teams

ISRO | *Research Intern* | *Semantic Segmentation of Remote Sensing Images*

Apr 2022 – Dec 2022

Objective: Develop an efficient CNN architecture using Tensorflow for semantic segmentation of geospatial images into 5 classes

- Modified standard U-Net architecture by adding **dilated & depthwise separable convolutions** to improve feature extraction
- Utilized a novel **unified focal loss** and created a **dataloader** for an **efficient input pipeline** to work with geospatial images

PROJECTS

Hospital Management System | *Database Management Systems Lab*

Feb 2023 – Mar 2023

Objective: Design and development of an extensible Hospital Management web application for Patient Care and Administration

- Developed a **Python Flask** based web application to connect **MySQL** to a **bootstrap front-end** coupled with **Jinja templates**
- Implemented **user session management** using **flask-login** & provided **access control** through **Python decorator functions**

Linux Shell Development | *Operating Systems Lab*

Jan 2023 – Feb 2023

Objective: Design a **pipeline-based** shell supporting **parallel command execution, detection of malware** and **file locks**

- Effectively managed **process groups** and monitored child processes using **signal handlers**, ensuring **synchronized execution**
- Implemented features such as **background execution, pipelining, wildcard handling, and command history navigation**

COMPETITIONS AND CONFERENCES

Inter IIT Tech Meet 12.0 - Gold Medalist | *Team Captain* | *DevRev – High Prep Event*

Sep 2023 – Dec 2023

Objective: Create a low-latency tool-use LLM which matches closed-source LLMs in performance while being cost-effective

- Created **RTaC** framework to convert tools to python functions, promoting **docstring-reading** capabilities in coding-base LLMs
- Utilized **PEFT** to finetune Code Llama 7B & DeepSeek 1.3B on 2500 examples of manually cleaned GPT generated synthetic data
- Achieved a competitive performance to GPT-4 at **20%** of the cost, supporting dynamic tool addition & mathematical/iterative logic

RELEVANT COURSEWORK

Theory + Lab: Operating Systems, Computer Networks, Database Management Systems, Computer Organisation & Architecture, Compilers, Software Engineering, Programming & Data Structures, Algorithms-I & II, Distributed Systems

Theory: Deep Learning, Machine Learning, Probability & Statistics, Statistical Inference, Discrete Structures, Linear Algebra

TECHNICAL SKILLS

Languages: C/C++, Python, LaTeX, SQL, Bash, MIPS, Assembly

Skills: Systems Programming, Socket Programming, Data Science, Object Oriented Design

Technologies/Frameworks: Keras, Tensorflow, NumPy, Pandas, Flask, scikit-learn, Git, C++ STL, C pthreads

EXTRACURRICULAR ACTIVITIES

Student Mentor | *Students' Welfare Group, IIT Kharagpur*

Dec 2022 – Present